Muhammad Ali Khan

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EDUCATION

University of Pécs — Pécs, Hungary
BSc Computer Science — *Sep. 2024 - Expected Jul. 2027*

Aalto University Summer School — Helsinki, Finland
Introduction to Artificial Intelligence and Machine Learning — *Jul. 2025 - Aug. 2025*

EXPERIENCE

Research Intern — May 2026 - Jul. 2026
Trustworthy Artificial Intelligence Laboratory (TAIL), University of Cagliari — *Cagliari, Italy*

- Conducted research on Retrieval-Augmented Generation (RAG) systems using the FlashRAG framework.
- Developed and evaluated retrieval pipelines, datasets, and benchmarks.
- Implemented the Mosaic-RAG microservice architecture using FastAPI and Docker.

AI & Machine Learning Summer School Participant — Jul. 2025 - Aug. 2025
Aalto University — *Helsinki, Finland*

- Completed intensive training in artificial intelligence and machine learning.
- Developed computer vision applications using YOLOv8.
- Worked on practical deep learning and data science projects.

LEADERSHIP & ACTIVITIES

Founding Team Member
Computer Science Student Association, University of Pécs — *Pécs, Hungary*

- Working with faculty support to establish a Computer Science student association.

Study Trip Coordinator
WATT Association, Saxion University of Applied Sciences — *Enschede, Netherlands*

- Coordinated study trips for WATT Association members.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, SQL, JavaScript, TypeScript
AI & ML: RAG, Computer Vision, Deep Learning, Large Language Models (LLMs), ONNX
Frameworks: PyTorch, YOLOv8, OpenCV, ONNX Runtime, LangChain, FAISS
Backend: FastAPI, Flask, REST APIs, Microservices
Systems: Linux, Raspberry Pi, Networking, Distributed Systems
Cloud & Tools: AWS (Solutions Architect Associate - In Progress), Docker, Git, GitHub Actions, WSL2
Simulation: MATLAB, Simulink, UAV Modelling, OpenRocket Simulator
Hardware & Design: Autodesk Fusion, Altium Designer, Quartus Prime, Tinkercad

PROJECTS

Traffic and Crowd Monitoring System [GitHub]

- BSc thesis project combining YOLOv8 detection, counting, analytics, and database-backed monitoring.

Smart Sonar Surveillance System [GitHub]

- Raspberry Pi-based ultrasonic scanning system with servo control, radar-style visualization, and alerts.

W65C02S-Based Computer

- Group Compilers and Assemblers project building a computer with a W65C02S CPU, W65C22 VIA, and AS7C256B SRAM.

Multi-UAV Path Planning [GitHub]

- Developing a MATLAB and Simulink simulation for 3D mapping, collision-free UAV routing, coordination, and task allocation.

Mosaic-RAG [GitHub]

- Modular RAG system with retrieval, generation, and evaluation services and FastAPI/Docker deployment.